MEDIUM · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

Two-variable data: Models and scatterplots

03

3 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

C

C. 0.22

Choice C is correct. A line in the xy-plane that passes through the points (x_1, y_1) and (x_2, y_2) has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes through the points with approximate coordinates $(0, 2.5)$ and $(10, 4.7)$. It follows that the slope of the line of best fit is approximately $\frac{4.7 - 2.5}{10 - 0}$, which is equivalent to $\frac{2.2}{10}$, or 0.22. Therefore, of the given choices, 0.22 is closest to the slope of the line of best fit.

Choice A is incorrect. This is closest to the negative y-coordinate of the y-intercept, not to the slope, of the line of best fit.

Choice B is incorrect. This is closest to the negative slope, not to the slope, of the line of best fit.

Choice D is incorrect. This is closest to the y-coordinate of the y-intercept, not to the slope, of the line of best fit.

Q2

C

C. 8.2

Choice C is correct. On the line of best fit, an x-value of 25.5 corresponds to a y-value between 8 and 8.5. Therefore, at $x = 25.5$, 8.2 is closest to the y-value predicted by the line of best fit.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q3

C

C. 2.4

Choice C is correct. At $x = 32$, the line of best fit has a y-value between 2 and 3. The only choice with a value between 2 and 3 is choice C.

Choice A is incorrect. This is the difference between the y-value predicted by the line of best fit and the actual y-value at $x = 32$ rather than the y-value predicted by the line of best fit at $x = 32$.

Choice B is incorrect. This is the y-value predicted by the line of best fit at $x = 31$ rather than at $x = 32$.

Choice D is incorrect. This is the y-value predicted by the line of best fit at $x = 33$ rather than at $x = 32$.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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